

AN EXOPLANET CODEX DOWNLOAD

Glossary of Terms

The words used to measure what a star is made of, in plain language

DOCUMENT	Glossary of Terms
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REFERENCE // WORDS USED ON THIS SITE

Glossary of *Terms*

65 terms, from absorption lines to uncertainty budgets. Each entry says what the word means, what it means in plain English, and where it is used in our Method.

A

Absorption line	
DEFINITION	A dark feature in a spectrum at the wavelength where atoms in the star's outer layers absorb light.
PLAIN ENGLISH	Each element absorbs only certain colours of light. Where it does, the star's spectrum has a dark notch. Each element has its own set of notches, so the notches tell us which elements are there.
UNITS	Wavelength in ångströms (Å)
EXAMPLE	Iron alone has thousands of lines in the Sun's visible spectrum. Asplund, Amarsi & Grevesse (2021) chose 40 Fe I and 13 Fe II lines from them for their solar iron value.
IN THE CODEX	Every abundance on this site comes from absorption lines. Method, section 01.
SEE ALSO	Spectrum, Equivalent width (EW), Blend

Abundance ratio, [X/H] and [X/Fe]

DEFINITION	[X/H] is a star's abundance of X minus the Sun's: [X/H] = $A(X)_{star} - A(X)_{Sun}$. [X/Fe] = [X/H] − [Fe/H] compares X with iron instead of hydrogen.
PLAIN ENGLISH	These say how a star differs from the Sun. Zero means the same as the Sun. +0.3 means about twice as much. [X/Fe] removes the star's overall metal content, so it shows which elements are unusual.
UNITS	dex
EXAMPLE	For the Sun every [X/H] and [X/Fe] is 0.00 by definition.
IN THE CODEX	Ratios like these are step 14 of the process, after the abundances themselves. Method, section 01.
SEE ALSO	Abundance, A(X), Metallicity, [Fe/H], Alpha enhancement, [α/Fe]

Abundance, A(X)

DEFINITION	The number of atoms of element X for every 10^{12} hydrogen atoms, on a log scale: $A(X) = \log_{10}(N_X/N_H) + 12$. Hydrogen is 12.00 by definition.
PLAIN ENGLISH	A way to say how much of an element a star has. Each step of 1 on the scale is a factor of ten. Iron at 7.46 means about 29 million iron atoms for every trillion hydrogen atoms.

UNITS	dex (logarithmic)
EXAMPLE	The Sun: A(Fe) = 7.46, A(O) = 8.69, A(C) = 8.46 (Asplund, Amarsi & Grevesse 2021).
IN THE CODEX	This is the number every Codex result reports. Method, section 06.
SEE ALSO	Abundance ratio, [X/H] and [X/Fe], dex, Metallicity, [Fe/H]

Alpha elements (α)

DEFINITION	Elements made mostly in massive stars and released quickly when those stars explode: O, Ne, Mg, Si, S, Ar, Ca and Ti. The name comes from how their nuclei are built, by adding helium nuclei (alpha particles), but the group is defined by where the elements come from, not by that alone.
PLAIN ENGLISH	A family of elements made in big, short-lived stars. Magnesium and silicon are among them, and they are the main rock-forming elements.

WHY THESE	Counting helium nuclei alone would include almost everything up to iron. Carbon-12 is three helium nuclei, and chromium-52 and even iron-56 are whole numbers of them too. So astronomers group elements by how they behave over the galaxy's history instead. The alpha elements rise above iron in old stars and fall back to the Sun's ratio in young ones, because they came early and iron came later. Carbon does not follow that pattern, since much of it comes from smaller stars late in their lives. Chromium is made alongside iron and keeps the same ratio to iron in stars of every age, so it counts as an iron-peak element. Titanium is not strictly built this way, but it follows the alpha pattern in real stars, so it is included.
UNITS	None (an element group)
EXAMPLE	Mg, Si, Ca and Ti are the alpha elements most often measured in Sun-like stars. Neon and argon cannot be measured in them at all: they leave no usable lines in the light from the star's surface.
IN THE CODEX	Silicon has a published solar result on this site. Method, section 01.
SEE ALSO	Alpha enhancement, [α/Fe], Iron-peak elements

Alpha enhancement, [α/Fe]

DEFINITION	The average of [X/Fe] over a set of alpha elements. It traces how a star's birth gas was enriched over the history of the galaxy. Studies differ on which elements go into the average; the most common choice is Mg, Si, Ca and Ti.
PLAIN ENGLISH	Early in the galaxy's life, massive stars made alpha elements quickly. Iron from slower kinds of supernova came later. Old stars therefore tend to have more alpha elements for their iron than young stars like the Sun.
UNITS	dex
EXAMPLE	The Sun: [α/Fe] = 0.00 by definition.
IN THE CODEX	Shown two ways, each worked out once all of its elements have passed review. **Accepted [α/Fe]** uses the usual four: Mg, Si, Ca and Ti. They are the alpha elements with the most clean lines in a Sun-like star. **Codex [α/Fe]** uses every alpha element we can measure: O, Mg, Si, S, Ca and Ti. Oxygen and sulfur are left out of the usual four because they have only a few lines, and oxygen's need large corrections. We measure both for the Sun anyway. Neon and argon are left out of both versions, because they leave no lines in the light from a Sun-like star. Method, section 01.
SEE ALSO	Alpha elements (α), Abundance ratio, [X/H] and [X/Fe]

Atlas (solar flux atlas)

DEFINITION	A very high-quality spectrum of the Sun's integrated light, published for other scientists to use. It is built from many observations and covers a wide wavelength range.
PLAIN ENGLISH	A reference photograph of sunlight, spread out by colour in great detail. Teams around the world measure from the same atlas, so their results can be compared directly.
UNITS	Wavelength in Å; normalised flux
EXAMPLE	The 1984 Kitt Peak solar atlas, and the IAG atlas from Göttingen (Reiners et al. 2016; Baker et al. 2020).
IN THE CODEX	Each atlas is a separate holding. The IAG atlas also serves as an independent check on continuum placement. Method, section 02.
SEE ALSO	Holding, Telluric absorption and correction, Continuum

ATLAS9

DEFINITION	A widely used grid of one-dimensional model atmospheres for stars like the Sun, computed by Robert Kurucz and updated by Castelli & Kurucz (2003). Not to be confused with a solar atlas.
PLAIN ENGLISH	A ready-made set of model star atmospheres, each one a table of temperature and pressure against depth.

UNITS	None (a model grid)
EXAMPLE	ATLAS9 and MARCS are both 1D. Running the same lines on each shows how much the answer depends on the model.
IN THE CODEX	One of two 1D model families we use. Swapping it for MARCS is one term in the error bar. Method, section 05.
SEE ALSO	Model atmosphere, LTE (local thermodynamic equilibrium)

B

Band	
DEFINITION	A wavelength range of the spectrum that is analysed under its own rules. The Codex uses six: near-UV, visible, red-optical, near-infrared, H and K.
PLAIN ENGLISH	We cut the spectrum into pieces because the problems change with colour. In the near-UV the lines crowd together. In the infrared the lines are sparse, but Earth's air absorbs strongly.
UNITS	Wavelength in Å
EXAMPLE	The visible band runs from 3780 to 6910 Å. The H band runs from 15007 to 17494 Å.

IN THE CODEX	Every band is measured on its own and never averaged with another. Method, section 02.
SEE ALSO	Holding, Orchestrator, Spectrum

Barycentric correction

DEFINITION	A shift applied to the wavelengths of an observation to remove the Earth's own motion around the Sun along the line of sight.
PLAIN ENGLISH	The Earth moves, and that motion slightly shifts the colours we see. The correction removes the shift so that only the star's motion is left.
UNITS	km/s, or the matching shift in Å
EXAMPLE	The standard HARPS data products already have this correction applied.
IN THE CODEX	Checked during the audit of each holding. Method, section 01.
SEE ALSO	Radial velocity (RV), Holding

Blend

DEFINITION	Two or more absorption lines so close in wavelength that they overlap in the spectrum.

PLAIN ENGLISH	Two notches that run into each other. If you measure one without allowing for the other, you get too much absorption and the wrong answer.
UNITS	None
EXAMPLE	The forbidden oxygen line at 6300.304 Å sits only 0.032 Å from a nickel line at 6300.336 Å.
IN THE CODEX	Blends are handled by calculating every known atomic and molecular line in the window together. Lines too blended to measure are left out, as published solar studies do. Method, section 04.
SEE ALSO	Spectrum synthesis, Absorption line, Spectral resolution, R

C

C/O ratio	
DEFINITION	The number of carbon atoms divided by the number of oxygen atoms in a star.
PLAIN ENGLISH	Carbon and oxygen compete for each other in the gas that forms planets. When oxygen wins, rocks are made of silicates, as on Earth. A star with much more carbon than usual could form very different planets.

UNITS	A ratio of numbers of atoms
EXAMPLE	The Sun: C/O = $10^{(8.46 - 8.69)} \approx 0.59$ (Asplund, Amarsi & Grevesse 2021).
IN THE CODEX	A derived ratio, worked out from the published carbon and oxygen results. Method, section 01.
SEE ALSO	Abundance, A(X), Mg/Si and Fe/Si ratios

CHNOPS

DEFINITION	The six elements that make up most of living matter: carbon, hydrogen, nitrogen, oxygen, phosphorus and sulfur.
PLAIN ENGLISH	The main ingredients of life as we know it. How much of each a star has hints at what its planets were built from.
UNITS	None (an element group)
EXAMPLE	Carbon, nitrogen and oxygen have published solar results on this site.
IN THE CODEX	Carbon, nitrogen and oxygen were among the first elements finished for the Sun. Method, section 01.
SEE ALSO	C/O ratio, Abundance, A(X)

Continuum

DEFINITION	The brightness a spectrum would have if there were no absorption lines. In a normalised spectrum it is set to 1.
PLAIN ENGLISH	The level the notches are measured down from. If the level is wrong, every notch looks too deep or too shallow.
UNITS	Normalised flux (1.0)
EXAMPLE	A line only 1% deep is changed noticeably by a continuum error of a tenth of a percent.
IN THE CODEX	We set the continuum near each line by comparing the observed spectrum with our calculated one, and repeat on a second atlas to price the error. Method, section 03.
SEE ALSO	Normalisation, Pseudo-continuum, Equivalent width (EW)

Correlated gf error

DEFINITION	An error in oscillator strengths that is shared by every line measured in the same laboratory study, so it does not shrink when more lines are averaged.
PLAIN ENGLISH	If a ruler is a millimetre short, measuring more things with it does not help. Lab gf values can be off in the same way: all lines from one study shift together.
UNITS	dex

EXAMPLE	Lab gf values often combine an atomic lifetime with how the decay splits between lines. An error in the lifetime moves all lines from that level together.
IN THE CODEX	For each gf source we record which part of its error is shared and which is per line. A source not yet reviewed is treated as fully shared. Method, section 07.
SEE ALSO	Oscillator strength, log gf, Laboratory gf, Uncertainty budget

Curve of growth

DEFINITION	The relation between a line's equivalent width and the abundance of the element. It has three parts: linear for weak lines, flat for saturated lines, and a damping part for very strong lines.
PLAIN ENGLISH	For a weak line, twice the atoms gives twice the absorption. Once the line's centre is nearly black, adding atoms barely changes it. That flat stretch is why strong lines tell us less.
UNITS	log(equivalent width / wavelength) against log abundance
EXAMPLE	Lines on the flat part depend strongly on microturbulence.
IN THE CODEX	Turns a measured equivalent width into an abundance in the profile-fit route. Method, section 06.

SEE ALSO	Equivalent width (EW), Saturated line, Microturbulence, ξ

D

Departure coefficient	
DEFINITION	The ratio of the true number of atoms in an energy level to the number LTE predicts. A value of 1 means LTE holds for that level.
PLAIN ENGLISH	A correction factor for each energy level of an atom. It says how far the real atoms are from the simple assumption, layer by layer through the atmosphere.
UNITS	Dimensionless ratio
EXAMPLE	Turbospectrum reads departure coefficients from an NLTE model atom and builds them into the calculated spectrum (Gerber et al. 2023).
IN THE CODEX	Used by the NLTE treatments that put NLTE inside the calculation. Method, section 05.
SEE ALSO	NLTE (non-local thermodynamic equilibrium), LTE (local thermodynamic equilibrium), Spectrum synthesis

dex

DEFINITION	Short for decimal exponent. A difference of 1 dex is a factor of ten.
PLAIN ENGLISH	The unit for log-scale numbers. Small differences are easier to think of as percentages: 0.01 dex is about 2%, and 0.1 dex is about 26%.
UNITS	Dimensionless (logarithmic)
EXAMPLE	[Fe/H] = +0.30 means about twice as much iron as the Sun, because $10^{0.30} \approx 2$.
IN THE CODEX	All abundances and error bars on this site are in dex. Method, section 05.
SEE ALSO	Abundance, A(X), Abundance ratio, [X/H] and [X/Fe]

E

Effective temperature, T_{eff} *T-eff*

DEFINITION	The temperature of a perfect black body that gives off the same total power per unit area as the star.
PLAIN ENGLISH	The star's surface temperature, in a precise sense. It sets which atoms are excited and so which lines are strong.

UNITS	Kelvin (K)
EXAMPLE	The Sun: 5772 K (Heiter et al. 2015).
IN THE CODEX	Fixed for the Sun before any line is measured. For the Sun it is known so well that its error-bar term comes out at zero. Method, section 07.
SEE ALSO	Surface gravity, log g, Microturbulence, ξ, Model atmosphere

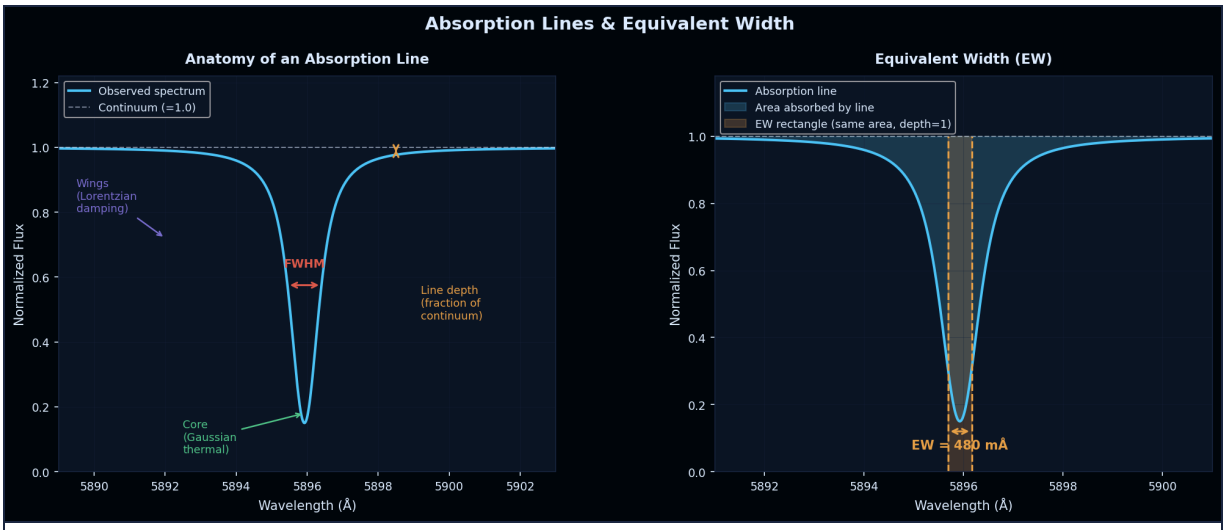
Engine

DEFINITION	One complete way of turning spectra into abundances, including how departures from LTE are handled. The Codex has two. One applies published NLTE corrections to each line's result. The other puts NLTE inside the spectrum calculation itself.
PLAIN ENGLISH	Two different calculators for the same question. If they agree, we trust the answer more. If they disagree, that tells us something about the physics.
UNITS	None
EXAMPLE	Iron is measured with published correction tables and with Turbospectrum's built-in NLTE, and both results are shown.
IN THE CODEX	Results from different engines and treatments are never averaged. Averaging would hide a disagreement that is real information. Method, section 05.

SEE ALSO	NLTE (non-local thermodynamic equilibrium), Orchestrator, Spectrum synthesis

Equivalent width (EW)

DEFINITION	The width of a completely dark strip with the same area as the absorption line.
PLAIN ENGLISH	One number for how much light a line removes, whatever its shape. A strong line has a large equivalent width and a faint line a small one.
UNITS	Ångströms (Å) or milliångströms (mÅ); 1 Å = 1000 mÅ
EXAMPLE	Many lines used for the Sun are a few tens of mÅ.
IN THE CODEX	Measured in the profile-fit route, in the visible and red bands where lines are well separated. Method, section 06.
SEE ALSO	Curve of growth, Absorption line, Continuum



Left: the parts of an absorption line. Right: the equivalent width is the width of a fully dark strip with the same area as the line.

F

Forbidden line

DEFINITION	A line from a transition so unlikely that it is rarely seen in the laboratory, but appears in stars because the light passes through so much gas. It is written in brackets, as in [O I].
PLAIN ENGLISH	A very weak kind of line. It forms under simple conditions, which makes it useful, but it is easily confused with any neighbour.
UNITS	None (a type of transition)
EXAMPLE	[O I] 6300.304 Å is a classic solar oxygen line. A nickel line sits 0.032 Å away.

IN THE CODEX	Used for oxygen, with its nickel neighbour included in the calculation. Method, section 04.
SEE ALSO	Blend, Oscillator strength, log gf

FWHM (full width at half maximum)

DEFINITION	The width of a line measured halfway between the continuum and the line's deepest point.
PLAIN ENGLISH	A simple measure of how wide a line is. Narrow lines are less likely to run into their neighbours.
UNITS	Ångströms (Å)
EXAMPLE	A line whose depth is 0.30 is measured for width where its depth is 0.15.
IN THE CODEX	Part of describing a line's shape in the profile-fit route. Method, section 06.
SEE ALSO	Equivalent width (EW), Spectral resolution, R, Voigt profile

G

Gaussian profile

DEFINITION	A bell-shaped line profile. It describes broadening by random gas motions and by the instrument.
PLAIN ENGLISH	The familiar bell curve. It drops off quickly away from the centre, so it has no long wings.
UNITS	Depth (0 to 1) and width σ (Å)
EXAMPLE	Weak lines in a Sun-like star look close to Gaussian.
IN THE CODEX	One part of the Voigt profile used in the profile-fit route. Method, section 06.
SEE ALSO	Voigt profile, Lorentzian profile, Thermal broadening

H

HARPS

DEFINITION	The High Accuracy Radial velocity Planet Searcher, a spectrograph on the ESO 3.6 m telescope at La Silla, Chile. Its resolution is about 115,000 and it covers 3780 to 6910 Å.

PLAIN ENGLISH	A very stable instrument built to find planets by the wobble they cause in their star. Its spectra are also sharp enough to measure elements.
UNITS	Resolution R ≈ 115,000
EXAMPLE	We use ten direct exposures of the Sun from HARPS, corrected for Earth's atmosphere with molecfit.
IN THE CODEX	One of the solar holdings in the visible band. Method, section 02.
SEE ALSO	Holding, Spectral resolution, R, Telluric absorption and correction

Holding

DEFINITION	One instrument's observations of one star, as stored and prepared by the Codex.
PLAIN ENGLISH	The set of spectra we have of a star from one instrument. The Sun has five: two versions of the Kitt Peak atlas, the IAG atlas, HARPS and CRIRES+.
UNITS	None
EXAMPLE	The CRIRES+ holding is infrared sunlight reflected off the asteroid Vesta.

IN THE CODEX	Each holding is corrected for Earth's atmosphere and normalised once, then frozen and used unchanged for every element. Method, section 02.
SEE ALSO	Band, Atlas (solar flux atlas), Telluric absorption and correction

HST/STIS and HST/COS

DEFINITION	Two ultraviolet spectrographs on the Hubble Space Telescope: the Space Telescope Imaging Spectrograph and the Cosmic Origins Spectrograph.
PLAIN ENGLISH	Earth's air blocks most ultraviolet light, so some lines can only be seen from space. These two instruments see them.
UNITS	None (instruments)
EXAMPLE	Some carbon, nitrogen, oxygen and silicon lines lie only in the ultraviolet.
IN THE CODEX	We hold Hubble spectra of planet hosts. They wait until the Sun is finished. Method, section 02.
SEE ALSO	MAST, Band

Hyperfine structure

DEFINITION	The splitting of one spectral line into several very close components, caused by the atomic nucleus. Different isotopes of an element split a line in a similar way.
PLAIN ENGLISH	What looks like one line is really a cluster of tiny lines. Treat it as one and the shape is wrong, so the answer is wrong.
UNITS	Component separations in mÅ
EXAMPLE	Manganese, cobalt, copper, scandium, vanadium, barium and europium lines need it.
IN THE CODEX	The calculation includes every component, but the gf is fixed once for the whole physical line and the components are scaled to match. Method, section 04.
SEE ALSO	Oscillator strength, log gf, Spectrum synthesis, Blend

I

Iron-peak elements

DEFINITION	Elements near iron in the periodic table, such as V, Cr, Mn, Fe, Co and Ni. Their nuclei are among the most tightly bound, so stars build up a lot of them.

PLAIN ENGLISH	Iron and its neighbours. They are made largely in supernova explosions and end up in planet cores.
UNITS	None (an element group)
EXAMPLE	Iron is the iron-peak element with the most usable lines.
IN THE CODEX	Iron is always measured first, as the check that everything before it works. Method, section 01.
SEE ALSO	Alpha elements (α), Metallicity, [Fe/H]

L

Laboratory gf	
DEFINITION	An oscillator strength measured in a laboratory, usually by timing how long an excited atom lives and measuring how its decay splits between lines.
PLAIN ENGLISH	The line strength measured directly from real atoms, not calculated. It comes with a stated error, which is what lets us put a number on how much it can be trusted.
UNITS	dex (as log gf)
EXAMPLE	Much of the solar iron work relies on laboratory gf from groups such as O'Brian et al. (1991).

IN THE CODEX	Any line with a laboratory gf is added to the Reference grade. Method, section 04.
SEE ALSO	Oscillator strength, log gf, Correlated gf error, Line grades: Reference, Codex and Deep

Line grades: Reference, Codex and Deep

DEFINITION	Three sets of lines, chosen by how well each line's oscillator strength is known. Reference: the lines and gf values of the published solar analysis, plus any line with a laboratory gf. Codex: any line whose gf has a published uncertainty, up to 60% deep. Deep: the same standard for lines deeper than 60%.
PLAIN ENGLISH	We first repeat the published work with the same lines, then widen the set. Every line in a grade has a gf error we can put a number on.
UNITS	None (a label)
EXAMPLE	A line whose gf has no stated uncertainty can still be measured, but it does not reach a grade.
IN THE CODEX	The grades are set before anything is measured. Method, section 04.
SEE ALSO	Published line set, Oscillator strength, log gf, NIST accuracy grade

Literature check

DEFINITION	A comparison of our result with the published solar values for the same element, each with the method it used.
PLAIN ENGLISH	Before we trust our number, we compare it with what other careful teams found and how they found it.
UNITS	dex
EXAMPLE	The visible band is the control. It has a known answer: iron at 7.46 (Asplund, Amarsi & Grevesse 2021).
IN THE CODEX	Published values are collected before measuring and compared after. The models are never adjusted to hit the published value. How far the method misses in the visible band goes into the error bar. Method, section 08.
SEE ALSO	Publication gate, Published line set, Uncertainty budget

Lorentzian profile

DEFINITION	A line profile with wide wings that fall off slowly away from the centre. It describes broadening by collisions in the gas.
PLAIN ENGLISH	Like a bell curve with long skirts. Strong lines show these wings clearly.
UNITS	Half-width γ (Å)

EXAMPLE	The broad wings of the sodium D lines are Lorentzian in shape.
IN THE CODEX	One part of the Voigt profile used in the profile-fit route. Method, section 06.
SEE ALSO	Voigt profile, Gaussian profile, Pressure broadening (damping)

LTE (local thermodynamic equilibrium)

DEFINITION	The assumption that in each layer of the atmosphere, the atoms' energy levels and ionisation are set by that layer's temperature alone.
PLAIN ENGLISH	The simplest way to calculate a spectrum. It treats every layer as if it were in a closed, evenly heated box.
UNITS	None (an approximation)
EXAMPLE	Most strong lines in the Sun form well enough in LTE to be a useful starting point. Many still need an NLTE correction.
IN THE CODEX	1D LTE is one of the treatments we run and report on its own. Method, section 05.
SEE ALSO	NLTE (non-local thermodynamic equilibrium), Departure coefficient, Model atmosphere

M

MAST	
DEFINITION	The Mikulski Archive for Space Telescopes, NASA's public archive of Hubble and other space-telescope data.
PLAIN ENGLISH	The library where Hubble's observations are kept for anyone to download.
UNITS	None (an archive)
EXAMPLE	Our Hubble spectra of planet hosts come from MAST.
IN THE CODEX	The source of our Hubble holdings. Method, section 02.
SEE ALSO	HST/STIS and HST/COS, Holding

Median over lines	
DEFINITION	The middle value of the line-by-line abundances, once they are put in order.
PLAIN ENGLISH	Each line gives its own answer. We take the middle one, so a single odd line cannot pull the result far.
UNITS	dex

EXAMPLE	Five lines giving 7.40, 7.44, 7.46, 7.47 and 7.90 have a median of 7.46. Their average would be 7.53.
IN THE CODEX	A result is the median over its lines. When the published analysis we are repeating used a plain average, we use the same. Method, section 06.
SEE ALSO	Abundance, A(X), Statistical and systematic uncertainty

Metallicity, [Fe/H]

DEFINITION	A star's iron abundance compared with the Sun's: [Fe/H] = $A(Fe)_{star} - A(Fe)_{Sun}$. It is used as a stand-in for all elements heavier than helium.
PLAIN ENGLISH	How metal-rich a star is compared with the Sun. Stars with more metals are more likely to host giant planets.
UNITS	dex
EXAMPLE	The Sun: [Fe/H] = 0.00 by definition.
IN THE CODEX	Iron is always measured first. For the Sun its metal-content error term comes out at zero. Method, section 07.
SEE ALSO	Abundance ratio, [X/H] and [X/Fe], Abundance, A(X), Iron-peak elements

Mg/Si and Fe/Si ratios

DEFINITION	The number of magnesium or iron atoms divided by the number of silicon atoms in a star.
PLAIN ENGLISH	Rocky planets are built mostly from magnesium, silicon, iron and oxygen. Mg/Si shapes which minerals form the mantle. Fe/Si hints at how big the iron core is.
UNITS	Ratios of numbers of atoms
EXAMPLE	The Sun: Mg/Si = $10^{(7.55 - 7.51)} \approx 1.10$ and Fe/Si = $10^{(7.46 - 7.51)} \approx 0.89$ (Asplund, Amarsi & Grevesse 2021).
IN THE CODEX	Derived ratios, worked out after the elements behind them are published. Method, section 01.
SEE ALSO	C/O ratio, Alpha elements (α)

Microturbulence, ξ *xi, "ksee"*

DEFINITION	A speed that stands in for small-scale gas motions in the atmosphere that are not included in a 1D model. It broadens the absorption of each atom.
PLAIN ENGLISH	Small swirls of gas spread a line's absorption over a wider range of colours. That matters most for strong lines, which are otherwise nearly black at their centre.
UNITS	km/s

EXAMPLE	The Sun: ξ = 1.0 km/s (Jofré et al. 2014).
IN THE CODEX	One of the stellar parameters fixed before measuring. Unlike temperature and gravity, its error term for the Sun is not zero, so we measure it by raising and lowering ξ. Method, section 07.
SEE ALSO	Saturated line, Curve of growth, Model atmosphere

Model atmosphere

DEFINITION	A table of temperature, pressure and density against depth through a star's outer layers. A 1D model uses flat layers. A full 3D model simulates the rising and sinking gas. A ⟨3D⟩ model is a 3D simulation averaged back into layers.
PLAIN ENGLISH	A recipe for what the star's outer layers look like, so we can predict how each line should form.
UNITS	None (a model)
EXAMPLE	We use two 1D families, MARCS (Gustafsson et al. 2008) and ATLAS9 (Castelli & Kurucz 2003), and ⟨3D⟩ models from the Stagger grid (Magic et al. 2013).
IN THE CODEX	Each atmosphere gives its own result. Swapping MARCS for ATLAS9 is one term in the error bar. Method, section 05.

SEE ALSO	ATLAS9, LTE (local thermodynamic equilibrium), NLTE (non-local thermodynamic equilibrium)

N

NIST accuracy grade

DEFINITION	A letter grade given by the US National Institute of Standards and Technology to an atomic transition probability, based on its estimated uncertainty. The scale runs AAA, AA, A+, A, B+, B, C+, C, D+, D and E.
PLAIN ENGLISH	A report card for a line's strength. Grade A means the value is good to a few percent. Grade C means within about 25%.
UNITS	Letter grade; A is within 3%, B within 10%, C within 25%, D within 50%
EXAMPLE	A line with a NIST grade of A to C can join the Codex grade. Grades D and E cannot.
IN THE CODEX	Used to decide which lines reach a grade. Method, section 04.
SEE ALSO	Oscillator strength, log gf, Line grades: Reference, Codex and Deep, Laboratory gf

NLTE (non-local thermodynamic equilibrium)

DEFINITION	Any calculation that does not assume LTE. The numbers of atoms in each energy level are worked out from all the processes that fill and empty them, including light coming up from deeper layers.
PLAIN ENGLISH	In a star's thin outer layers, light from below also excites the atoms. Allowing for this changes how deep a line is predicted to be.
UNITS	dex (as a correction to an abundance)
EXAMPLE	The oxygen lines near 7774 Å are a well-known case where NLTE matters a great deal.
IN THE CODEX	We include NLTE two ways: published correction tables for each line, and departure coefficients inside the calculation. Each is its own result. Method, section 05.
SEE ALSO	LTE (local thermodynamic equilibrium), Departure coefficient, Engine

Normalisation

DEFINITION	Dividing a spectrum by its continuum so that the continuum is at 1. Line depths are then fractions of the continuum.
PLAIN ENGLISH	Levelling the spectrum so every notch is measured from the same height.

UNITS	None (flux becomes a fraction)
EXAMPLE	Near each line we compare the observed spectrum with our calculated one and set the level from that comparison. Points that disagree badly, such as an unlisted line, are clipped out first.
IN THE CODEX	Done once per holding with Earth's atmosphere removed, then frozen for every element. Method, section 03.
SEE ALSO	Continuum, Pseudo-continuum, Telluric absorption and correction

O

Orchestrator

DEFINITION	The program that runs the whole process for one star and one element from a single command. It works through every combination of band, holding and treatment, and reports the status of each one.
PLAIN ENGLISH	One command per element. The program does the measuring and tells us, for every combination, whether it is done, held back and why, or not run yet.
UNITS	None

EXAMPLE	For the Sun, one command per element measures every band, instrument and treatment and builds each error bar.
IN THE CODEX	People approve line removals and publication; the orchestrator does the rest. Method, section 01 and the Science Architecture.
SEE ALSO	Engine, Band, Publication gate

Oscillator strength, log gf

DEFINITION	A measure of how strongly an atom absorbs at one line, written as the log of g × f: the statistical weight of the lower level times the oscillator strength.
PLAIN ENGLISH	How good an atom is at absorbing that particular colour. A wrong value gives a wrong abundance, however good the spectrum is.
UNITS	Dimensionless (logarithmic)
EXAMPLE	Fe I 5247.05 Å has log gf = −4.961 in the published solar iron set (Asplund, Amarsi & Grevesse 2021).
IN THE CODEX	Fixed once for each physical line from a single master table. A line with no entry stops the run. Method, section 04.
SEE ALSO	Laboratory gf, NIST accuracy grade, Correlated gf error

P

Pressure broadening (damping)

DEFINITION	Widening of a line by collisions between the absorbing atom and other particles in the gas. It gives a line Lorentzian wings.
PLAIN ENGLISH	Atoms that bump into neighbours absorb a slightly wider range of colours. In dense gas, strong lines get wide wings.
UNITS	Å; described by damping constants
EXAMPLE	The sodium D lines in the Sun have wings that stretch for several ångströms.
IN THE CODEX	Part of every calculated line profile. Method, section 06.
SEE ALSO	Lorentzian profile, Voigt profile, Saturated line

Pseudo-continuum

DEFINITION	The highest level a spectrum reaches where lines cover every wavelength, so the true continuum is never seen.
PLAIN ENGLISH	In the near-UV the notches overlap everywhere. The best we can do is a level close to, but below, the true one.
UNITS	Normalised flux

EXAMPLE	Below about 3780 Å no solar line stands alone.
IN THE CODEX	Its uncertainty is priced as a separate term, for near-UV results only. Method, section 03.
SEE ALSO	Continuum, Normalisation, Band

Publication gate

DEFINITION	The checks a result must pass before it is published: every error-bar term present, a comparison with the literature, and a person's approval.
PLAIN ENGLISH	Being computed is not enough. A result goes on the site only when it has a complete error bar and someone has reviewed it.
UNITS	None
EXAMPLE	A result missing any error-bar term is held back and no error bar is published for it. A held result is one we know we still owe, with the reason recorded.
IN THE CODEX	Every proposed line removal and every publication is approved by a person. Method, section 08.
SEE ALSO	Literature check, Uncertainty budget, Superseded result

Published line set

DEFINITION	The lines, and their gf values, that a published solar analysis used for an element.
PLAIN ENGLISH	The exact recipe another team used. We repeat it first, so that any difference from their answer comes from our spectra or our method, and we can find it.
UNITS	None
EXAMPLE	For iron, Asplund, Amarsi & Grevesse (2021) used 40 Fe I and 13 Fe II lines. For other elements they cite the paper the lines come from.
IN THE CODEX	The published set, plus any line with a laboratory gf, makes the Reference grade. Method, section 04.
SEE ALSO	Line grades: Reference, Codex and Deep, Literature check, Oscillator strength, log gf

R

Radial velocity (RV)

DEFINITION	A star's speed toward or away from us, measured from the shift of its spectral lines.

PLAIN ENGLISH	A star moving away looks slightly redder, and one moving toward us slightly bluer. The lines must be shifted back to their rest wavelengths before measuring.
UNITS	km/s
EXAMPLE	At 5900 Å, a speed of 30 km/s shifts a line by about 0.6 Å.
IN THE CODEX	Each holding is checked to serve every line at its rest wavelength. Method, section 02.
SEE ALSO	Barycentric correction, Holding

Rotational broadening, v sin i

DEFINITION	Widening of lines by the star's spin. One edge of the star moves toward us and the other away.
PLAIN ENGLISH	A spinning star smears its lines. Slow spinners like the Sun have sharp lines.
UNITS	km/s
EXAMPLE	The Sun: v sin i = 1.8 km/s.
IN THE CODEX	Part of the line broadening used in every calculated spectrum. Method, section 06.
SEE ALSO	Thermal broadening, Pressure broadening (damping)

S

s-process and r-process elements

DEFINITION	Elements heavier than iron made by capturing neutrons. The slow (s) process works inside aging giant stars. The rapid (r) process needs extreme events such as merging neutron stars.
PLAIN ENGLISH	Two very different ways nature builds heavy elements. Barium and yttrium come mostly from the slow route. Europium comes mostly from the rapid one.
UNITS	None (element groups)
EXAMPLE	Ba II, Y II and Eu II lines are visible in Sun-like stars.
IN THE CODEX	Heavy elements will run through the same process after the lighter ones. Method, section 01.
SEE ALSO	Hyperfine structure, Abundance, A(X)

Saturated line

DEFINITION	A line so strong that its centre is nearly black, so its depth barely changes when the abundance changes.
PLAIN ENGLISH	Once the notch is almost fully dark, adding atoms hardly changes it. The line then tells us more about gas motions than about how many atoms there are.

UNITS	None
EXAMPLE	The sodium D lines in the Sun are saturated.
IN THE CODEX	A saturated line that the published solar analyses did not use is proposed for removal, and a person approves each one. Method, section 04.
SEE ALSO	Curve of growth, Microturbulence, ξ, Line grades: Reference, Codex and Deep

Signal-to-noise ratio, S/N

DEFINITION	The strength of the signal divided by the size of the random noise in a spectrum.
PLAIN ENGLISH	How clean the spectrum is. With high S/N, faint notches stand out clearly from the noise.
UNITS	Dimensionless ratio
EXAMPLE	Combining N similar exposures raises S/N by about $\sqrt{N}$.
IN THE CODEX	Noise in the spectrum feeds the statistical part of each error bar. Method, section 07.
SEE ALSO	Statistical and systematic uncertainty, Spectrum

Spectral resolution, R

DEFINITION	R = λ/Δλ, where Δλ is the smallest wavelength gap the instrument can separate at wavelength λ.
PLAIN ENGLISH	How fine the detail is. At higher R, two lines close together appear as two lines, not one smudge.
UNITS	Dimensionless ratio
EXAMPLE	HARPS has R ≈ 115,000, so at 6000 Å it separates features about 0.05 Å apart. The 2010 thesis used ELODIE, at R ≈ 42,000.
IN THE CODEX	Every holding we use has high enough resolution to separate most solar lines. Method, section 02.
SEE ALSO	HARPS, Blend, FWHM (full width at half maximum)

Spectral type

DEFINITION	A classification of stars by surface temperature and the pattern of their lines, from hottest to coolest: O, B, A, F, G, K, M.
PLAIN ENGLISH	A quick label for what kind of star it is. The Sun is a G star. Slightly cooler, orange stars are K stars.
UNITS	None (a classification)

EXAMPLE	The Sun: G2V. 55 Cancri A, a planet host we hold spectra for, is a K star.
IN THE CODEX	The method is proven on the Sun before other stars. Method, section 02.
SEE ALSO	Effective temperature, T_{eff}, Surface gravity, log g

Spectrum

DEFINITION	The light from an object spread out by wavelength, showing how much light arrives at each wavelength.
PLAIN ENGLISH	A star's light spread into a rainbow, in great detail. The dark notches in it are the star's chemical fingerprint.
UNITS	Wavelength in Å; normalised flux from 0 to 1
EXAMPLE	The solar spectra we use run from 3000 Å in the near-UV to 24846 Å in the infrared.
IN THE CODEX	Every result starts from a spectrum that has been corrected and normalised. Method, section 01.
SEE ALSO	Absorption line, Continuum, Spectral resolution, R

Spectrum synthesis

DEFINITION	Calculating the spectrum a model atmosphere would produce for a given set of abundances, including every known atomic and molecular line in the window.
PLAIN ENGLISH	We compute what the spectrum should look like, then change the amount of the element until the computed spectrum matches the real one.
UNITS	None (a method)
EXAMPLE	We use Turbospectrum (Plez 2012) through iSpec (Blanco-Cuaresma et al. 2014).
IN THE CODEX	The main route. In the near-UV it is the only route allowed, because no line there stands alone. Method, section 06.
SEE ALSO	Blend, Model atmosphere, Engine

Statistical and systematic uncertainty

DEFINITION	Statistical uncertainty comes from the scatter between lines. Systematic uncertainty is how far the answer moves when a choice we had to make is changed.
PLAIN ENGLISH	Statistical error shrinks when more lines are used. Systematic error does not, because every line shares the same choice.
UNITS	dex

EXAMPLE	With several lines, the statistical error is the scatter divided by the square root of the number of lines.
IN THE CODEX	Both parts go into every published error bar. Method, section 07.
SEE ALSO	Uncertainty budget, Correlated gf error, Median over lines

Superseded result

DEFINITION	A published result that has been replaced by a newer measurement of the same thing.
PLAIN ENGLISH	When we redo a measurement, for example after a fix to the atomic data, the new result replaces the old one. The old one is not deleted.
UNITS	None
EXAMPLE	A superseded result is moved to an archive with the record of why it was replaced.
IN THE CODEX	Only current results are shown, and every earlier number can still be traced. Method, section 08.
SEE ALSO	Publication gate, Literature check

Surface gravity, log g

DEFINITION	The base-10 logarithm of the gravity at the star's surface, in cgs units (cm/s²). It sets the pressure in the atmosphere.
PLAIN ENGLISH	How strongly the star pulls on its outer layers. Higher gravity means denser gas, which changes line widths and how many atoms are ionised.
UNITS	$\log_{10}$(cm/s²)
EXAMPLE	The Sun: log g = 4.44 (Heiter et al. 2015).
IN THE CODEX	Fixed before measuring. For the Sun its error-bar term comes out at zero. Method, section 07.
SEE ALSO	Effective temperature, T_{eff}, Model atmosphere

T

Telluric absorption and correction

DEFINITION	Telluric absorption is absorption of starlight by Earth's own atmosphere, mostly water vapour and oxygen. Telluric correction removes it from the spectrum.

PLAIN ENGLISH	Before starlight reaches a telescope on the ground, Earth's air takes its own bites out of it. Those bites land on top of the star's lines and must be removed first.
UNITS	None
EXAMPLE	For HARPS and CRIRES+ we model the air on the night of each observation with molecfit (Smette et al. 2015).
IN THE CODEX	Every spectrum taken from the ground is corrected. Only a telescope in space escapes this. Method, section 03.
SEE ALSO	Holding, Atlas (solar flux atlas), Normalisation

Thermal broadening

DEFINITION	Widening of a line by the random motion of the absorbing atoms, which Doppler-shifts each atom's absorption slightly.
PLAIN ENGLISH	Hot atoms move fast in all directions. Each one sees the light at a slightly different colour, which spreads the line.
UNITS	Å; gives a Gaussian shape
EXAMPLE	Lighter atoms move faster at the same temperature, so their lines are wider.
IN THE CODEX	Part of every calculated line profile. Method, section 06.

SEE ALSO	Gaussian profile, Microturbulence, ξ

U

Uncertainty budget	
DEFINITION	The list of every source of error in a result, each measured separately, then added in quadrature: the square root of the sum of squares.
PLAIN ENGLISH	A full account of what could move the answer. For each choice we re-run the same lines with that choice changed and record how far the answer moves.
UNITS	dex
EXAMPLE	Terms include the gf values, temperature, gravity, metal content, microturbulence, continuum, fit window, Earth's atmosphere, NLTE, model atmosphere, hyperfine structure, blends and a method check.
IN THE CODEX	Every result carries every term. If one is missing, the result is held back. Method, section 07.
SEE ALSO	Statistical and systematic uncertainty, Correlated gf error, Publication gate

V

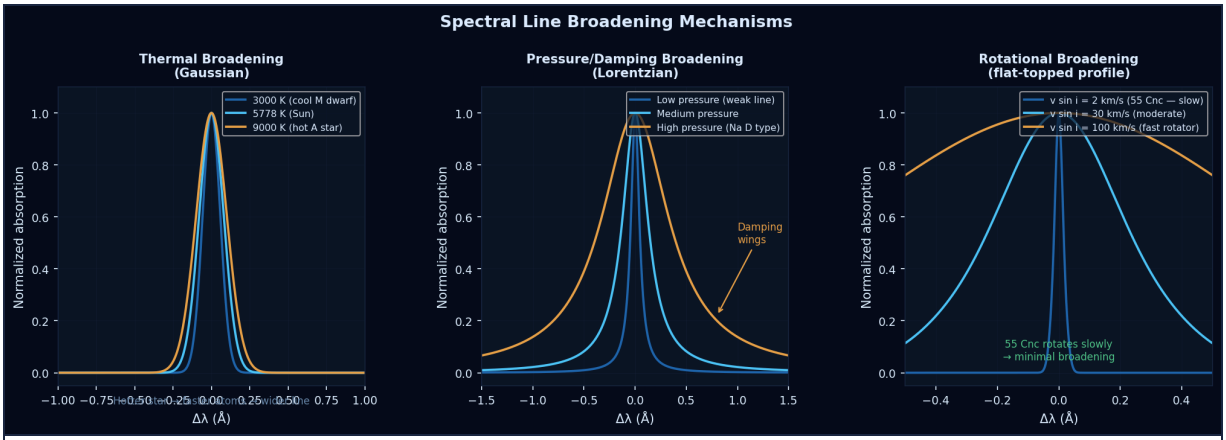
VALD

DEFINITION	The Vienna Atomic Line Database, a widely used collection of atomic and molecular line data: wavelengths, energy levels, oscillator strengths and broadening constants (Ryabchikova et al. 2015).
PLAIN ENGLISH	A large public catalogue of spectral lines. It tells us where lines are and roughly how strong, but not every value in it has a measured error.
UNITS	None (a database)
EXAMPLE	We extract solar line lists from VALD3, with hyperfine components included.
IN THE CODEX	Supplies the neighbouring lines in each synthesis window. The gf of each line we measure comes from the master table instead. Method, section 04.
SEE ALSO	Oscillator strength, log gf, Spectrum synthesis, Blend

Voigt profile

DEFINITION	A line profile that combines a Gaussian core with Lorentzian wings: the mathematical convolution of the two.

PLAIN ENGLISH	The realistic shape of a stellar line: bell-shaped in the middle, with longer wings for strong lines.
UNITS	Centre (Å), Gaussian width and Lorentzian width (Å)
EXAMPLE	Fitting a strong line with only a Gaussian misses its wings and underestimates its equivalent width.
IN THE CODEX	The shape fitted in the profile-fit route. Method, section 06.
SEE ALSO	Gaussian profile, Lorentzian profile, Equivalent width (EW)



Three ways a line is widened. Left: thermal motion gives a bell-shaped core. Centre: collisions give long wings. Right: a fast-spinning star smears the line flat; a slow spinner like the Sun barely does.